Jackie (Jingkai) Shao

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Work Experience

Manufacturing Controls Engineer — **Filament Winding | RAPID | KUKA Robot Lauguage (KRL)**
OPmobility (Plastic Omnium) | Internship — *April 2026 – Present*

Mechanical Engineer — **SolidWorks | AutoCAD | Ansys | Triflex | Welding**
Chart Industries, Inc. | Co-op — *December 2025 – March 2026*

- Designed 10+ SolidWorks 3-D models using DFM/DFA and AutoCAD 2-D drawings (Part and Assembly) for the bulk storage tanks, improving the First Pass Yield (FPY) from 78% to 85%.
- Conducted a pipeline stress analysis using Triflex and added expansion loops or bellows (design for strength and stiffness) to prevent high concentrations of stress, achieving the $3.5 - 4.0$ safety factor outlined by ASME Section VIII.
- Conducted a Conjugate Heat Transfer Analysis in Ansys Fluent to simulate the heat transfer across the fluid boundary layers near the internal tank wall, resulting in a 4% decrease of Nusselt Number (Nu).
- Developed MIG/TIG welding procedures for external tank structures, redesigning joint geometries to improve weld accessibility and reduce weld time per assembly by 10%.

System Design & Automation Engineer — **CATIA | Abaqus | KUKA.Sim | PLC Programming**
OPmobility (Plastic Omnium) | Internship — *May 2025 – August 2025*

- Designed 5+ CATIA 3D models using DFM/DFA and 2D drawings (Part and Assembly) with GD&T for thermoplastic liners used in Type IV hydrogen mobility vessels, resulting in better compatibility with automated manufacturing.
- Performed non-linear FEA in Abaqus using raw data from proprietary ComposiCAD software to pinpoint structural weaknesses, predict burst pressure, and simulate pressure cycle testing, reducing prototyping iteration time by 30%.
- Developed a Digital Twin Robotic Arm in KUKA.Sim to perform a Gage R&R study on fibre winding angle, improving the measurement accuracy of the rotary encoders by 10%.
- Conducted PLC programming on the Siemens S7-1500 to reduce the wedge effect of the composite filament layers by using a Tension Profile, improving manufacturing consistency from $C_{pk} < C_p$ to $Cpk \geq 1.33$.

Projects

Tooling Design (Jigs & Fixtures) | *Github* — **Associated with UWaterloo Engineering**

- Designed a jig with spring-loaded hold-down clamps (DFA) in SolidWorks, eliminating the need for threaded fasteners, which reduces the part assembly time by 20%.
- Designed a fixture with single-motion clamping instead of a screw-down clamp (DFM), eliminating the need for precision-threaded holes in the fixture body, which reduces manufacturing cost by 10%.

LNG Cryogenic Storage Tank Modeling | *Github* — **Associated with Chart Industries**

- Designed a 3-D Modeling of LNG Cryogenic Storage Tank in SolidWorks with instrumentation panels (HMI + gauges) and piping functions (valves + regulators + reliefs), providing clear visualization to clients and technicians.

Filament Winding Robotic Arm Digital Twin (KUKA.Sim) | *Github* — **Associated with OPmobility**

Education

B.A.Sc in Mechanical Engineering — **September 2025 - June 2030**
University of Waterloo, Honours Program with Co-op — *Ontario, Canada*

- **Relevant Coursework:** Engineering Design, C++ Programming, Material Science, Electrical Circuits, Mechanics, CFD/FEA Workshop

Skills

Software: CATIA, SolidWorks, Abaqus, Ansys, AutoCAD, Triflex, Robotstudio, KUKA.Sim, Microsoft/Google Workspace
Technical: FEA, Robotstudio, KUKA.Sim, GD&T, Pipeline Stress Analysis, Heat Transfer & Thermal Modeling
Quality & Statistics: Process Capability (C_p/C_{pk}), Gage Repeatability and Reproducibility (GR&R), Lean Six Sigma
Manufacturing: CNC Machining, Manual Lathe/Mill, Filament Winding, Blow/Injection Moulding, Welding
Programming: RAPID, KRL, C++, Python, MATLAB, PLC Programming

Licenses & Certifications

Fundamentals of CATIA V-5 | *Certificate* — **Issued by Coursera**

Finite Element Analysis Convergence and Mesh Independence | *Certificate* — **Issued by Coursera**

FEM - Linear, Nonlinear Analysis & Post-Processing | *Certificate* — **Issued by Coursera**

Private Pilot Licenses - Aeroplanes | *License* — **Issued by Transport Canada**